Simulation of different scenarios

Setup

```
library(here)
library(tidyverse)
library(data.table)
library(tidybayes)
library(rmsb)
library(rms)
library(Hmisc)
library(arrow)
library(VGAM)
library(patchwork)

set.seed(1234)
```

Load model and dataset

```
df <- read_parquet(here("data", "censored_qol.parquet")) |>
  mutate(
    ecog_fstcnt = if_else(ecog_fstcnt == "4" | ecog_fstcnt == "3", "3plus",
ecog_fstcnt),
    ecog_fstcnt = factor(ecog_fstcnt, ordered = FALSE)
  )

model <- readRDS(here("1-qol", "models", "qol_long_ord_pol.rds"))
model
```

```
Warning in blrmStats(x, ns = ns): some observations are censored but only left
endpoint used in summary statistics
```

```
Bayesian Constrained Partial Proportional Odds Ordinal Logistic Model

Dirichlet Priors With Concentration Parameter 0.278 for Intercepts

blrm(formula = Ocens(y.a, y.b) ~ tx + pol(week, 2) + yprev *
    gap + ecog_fstcnt + diagnosis + pat_age + gender + plan_fstcnt_coded,
    ppo = ~week, cppo = function(y) y, data = df, iter = 2000,
    chains = 4, refresh = 5, method = "sampling")


Frequencies of Responses

   1    2    3    4    5    6    7    8
5388   60   82   55   45   10    8   77

                    Mixed Calibration/             Discrimination              Rank
Discrim.
```

```
                 Discrimination Indexes                    Indexes
Indexes
    Obs 5725    B 0.216 [0.173, 0.254]    g  1.316 [1.088, 1.508]    C   0.613 [0.591,
0.636]
Censored5366                              gp 0.257 [0.226, 0.284]    Dxy 0.227 [0.183,
0.273]
      L=5366                              EV 0.214 [0.161,
0.251]
   Draws4000                              v  1.405 [0.956,
1.807]
    Chains 4                              vp  0.052 [0.04,
0.062]

Time355.4s
    p
28

                     Mean Beta Median Beta S.E.   Lower   Upper   Pr(Beta>0)
y>=2                  2.1784    2.2044     0.9396  0.3681  3.9978 0.9905
y>=3                  0.4585    0.4700     0.9084 -1.2513  2.2657 0.6968
y>=4                 -0.8616   -0.8513     0.9110 -2.6188  0.8934 0.1752
y>=5                 -1.8126   -1.7988     0.9279 -3.5787 -0.0168 0.0198
y>=6                 -3.2190   -3.2099     0.9497 -5.0323 -1.3788 0.0000
y>=7                 -3.8980   -3.9033     0.9712 -5.7841 -2.0415 0.0000
y>=8                 -5.1713   -5.1648     0.9917 -7.1632 -3.3062 0.0000
tx=1                 -0.2718   -0.2738     0.1686 -0.5906  0.0590 0.0542
week                 -0.0303   -0.0299     0.0540 -0.1437  0.0708 0.2848
week^2               -0.0009   -0.0008     0.0017 -0.0042  0.0025 0.3125
yprev=2              -0.2708   -0.2940     0.8514 -1.9563  1.3469 0.3658
yprev=3              -0.8154   -0.8507     0.8192 -2.4644  0.7847 0.1600
yprev=4               0.2452    0.2085     0.7823 -1.2395  1.7978 0.6070
yprev=5              -0.9311   -0.9599     0.8223 -2.5532  0.6543 0.1322
yprev=6              -0.8087   -0.8347     0.8349 -2.4314  0.8762 0.1650
yprev=7               0.3561    0.3402     0.9356 -1.5863  2.1349 0.6500
gap                   0.1072    0.1064     0.0737 -0.0340  0.2495 0.9312
ecog_fstcnt=2         1.3319    1.3347     0.2330  0.8830  1.7957 1.0000
ecog_fstcnt=3plus     2.6632    2.6654     0.3528  1.9996  3.3724 1.0000
diagnosis=2           0.0013   -0.0018     0.3301 -0.6156  0.6639 0.4980
diagnosis=3          -0.7023   -0.6973     0.3595 -1.4303 -0.0275 0.0238
diagnosis=4          -0.5999   -0.5942     0.4269 -1.4681  0.1889 0.0765
diagnosis=5          -0.1185   -0.1197     0.2216 -0.5777  0.2772 0.2998
pat_age               0.0154    0.0154     0.0066  0.0026  0.0281 0.9878
gender=male           0.1180    0.1192     0.1776 -0.2209  0.4627 0.7498
plan_fstcnt_coded=2 -0.0747   -0.0712     0.2947 -0.6398  0.5004 0.3992
plan_fstcnt_coded=3 -0.1922   -0.1941     0.2494 -0.6728  0.3053 0.2120
plan_fstcnt_coded=4  0.4074    0.4111     0.3182 -0.2055  1.0197 0.9002
plan_fstcnt_coded=5 -0.2718   -0.2644     0.3112 -0.8696  0.3464 0.1862
yprev=2 * gap        -0.0762   -0.0751     0.0817 -0.2393  0.0792 0.1782
yprev=3 * gap         0.0557    0.0559     0.0764 -0.0941  0.2097 0.7742
yprev=4 * gap        -0.0171   -0.0153     0.0753 -0.1626  0.1338 0.4188
yprev=5 * gap         0.0509    0.0517     0.0774 -0.1028  0.2033 0.7498
yprev=6 * gap        -0.0075   -0.0070     0.0766 -0.1558  0.1426 0.4645
yprev=7 * gap        -0.1286   -0.1270     0.0905 -0.3150  0.0417 0.0752
week x f(y)          -0.0062   -0.0062     0.0067 -0.0200  0.0066 0.1760
                     Symmetry
```

```
y>=2                0.94
y>=3                0.96
y>=4                0.94
y>=5                0.93
y>=6                0.96
y>=7                0.94
y>=8                0.96
tx=1                1.03
week                0.99
week^2              1.00
yprev=2             1.12
yprev=3             1.11
yprev=4             1.09
yprev=5             1.11
yprev=6             1.07
yprev=7             1.04
gap                 1.03
ecog_fstcnt=2       1.00
ecog_fstcnt=3plus   0.99
diagnosis=2         0.99
diagnosis=3         0.95
diagnosis=4         0.96
diagnosis=5         1.00
pat_age             1.01
gender=male         1.02
plan_fstcnt_coded=2 1.01
plan_fstcnt_coded=3 1.02
plan_fstcnt_coded=4 1.03
plan_fstcnt_coded=5 0.97
yprev=2 * gap       0.97
yprev=3 * gap       0.96
yprev=4 * gap       0.98
yprev=5 * gap       0.95
yprev=6 * gap       0.95
yprev=7 * gap       0.97
week x f(y)         1.01
```

Extract median of posterior draws

```
draws <- as.data.table(model$draws)

coef <- draws |>
  summarise_all(median)

coef
```

```
      y>=2      y>=3       y>=4      y>=5      y>=6      y>=7      y>=8
1 2.204351 0.4699935 -0.8512987 -1.798756 -3.209936 -3.903308 -5.164819
        tx=1        week       week^2    yprev=2    yprev=3   yprev=4
1 -0.2738061 -0.02993684 -0.000844602 -0.2939562 -0.8506924 0.2085366
     yprev=5    yprev=6    yprev=7       gap ecog_fstcnt=2 ecog_fstcnt=3plus
1 -0.9598782 -0.834665 0.3402182 0.1063974      1.334701          2.665404
   diagnosis=2 diagnosis=3 diagnosis=4 diagnosis=5    pat_age gender=male
```

```
1 -0.001814473  -0.6973488  -0.5941715  -0.1197462 0.01535362   0.1192489
  plan_fstcnt_coded=2 plan_fstcnt_coded=3 plan_fstcnt_coded=4
1         -0.07120574          -0.1940889           0.4110754
  plan_fstcnt_coded=5 yprev=2 * gap yprev=3 * gap yprev=4 * gap yprev=5 * gap
1          -0.2644251   -0.07513555    0.05590378   -0.01534478     0.0516682
  yprev=6 * gap yprev=7 * gap  week x f(y)
1  -0.006963717      -0.12704 -0.006221283
```

Baseline-Covariates

```
N <- 100000 # Simulation size

baseline_w <- df |>
  select(-c(y, y.a, y.b, week, surv_weeks, tx, gap)) |>
  group_by(pat_id) |>
  slice_head(n = 1) |>
  ungroup()

i <- sample(1:nrow(baseline_w), N, replace = TRUE)
sim_baseline_data <- baseline_w[i, ] |>
  mutate(pat_id = row_number())

init <- sim_baseline_data$yprev # initial states to be passed to g function

baseline_formula <- ~ ecog_fstcnt + diagnosis + gender + pat_age + plan_fstcnt_coded
X_baseline <- model.matrix(baseline_formula, data = sim_baseline_data)[, -1] # Baseline
values dummy coded, remove intercept

tx <- sample(0:1, N, replace = TRUE) # Sample treatment assignment
X <- cbind(X_baseline, tx)
```

Simulation Functions

Function A

- Proportional Part: Effects that are constant across all k-1 logits (e.g., yprev, gap, baseline covariates, pol(week,2)).
- Non-Proportional Part: The week x f(y) term, which varies with the outcome level k. For our model cppo = function(y) y, this term is k * week * _{week}.

```
ints <- unlist(coef[1:7])
extra <- unlist(coef[- (1:8)]) #remove tx coef as well

names(extra)[names(extra) == "ecog_fstcnt=2"] <- "ecog_fstcnt2"
names(extra)[names(extra) == "ecog_fstcnt=3plus"] <- "ecog_fstcnt3plus"
names(extra)[names(extra) == "diagnosis=2"] <- "diagnosis2"
names(extra)[names(extra) == "diagnosis=3"] <- "diagnosis3"
names(extra)[names(extra) == "diagnosis=4"] <- "diagnosis4"
names(extra)[names(extra) == "diagnosis=5"] <- "diagnosis5"
names(extra)[names(extra) == "plan_fstcnt_coded=2"] <- "plan_fstcnt_coded2"
names(extra)[names(extra) == "plan_fstcnt_coded=3"] <- "plan_fstcnt_coded3"
names(extra)[names(extra) == "plan_fstcnt_coded=4"] <- "plan_fstcnt_coded4"
names(extra)[names(extra) == "plan_fstcnt_coded=5"] <- "plan_fstcnt_coded5"
```

```r
names(extra)[names(extra) == "gender=male"] <- "gendermale"

g <- function(yprev, t, gap, X, parameter = 0, extra) {

  # X is a vector for one patient, containing baseline covariates and tx (which we want
to remove)
  # yprev is the previous state (character)
  # t is the current week (numeric)
  # gap is assumed to be 1 for this simulation
  gap <- 1

  # 0. Treatment Effect
  # The treatment indicator 'tx' is the last element of the X vector.
  tx_value <- X[length(X)]
  tx_effect <- parameter * tx_value

  # 1. Baseline effect
  # X contains the non-treatment variables. We match its names to the 'extra' vector.
  X_nontx <- X[-length(X)] # X is a single vector. We remove tx which is as the end of
this vector
  baseline_coef_names <- names(X_nontx)

  baseline_effect <- sum(extra[baseline_coef_names] * X_nontx)

  # 2.  time effect (proportional part) from pol(week, 2)
  time_effect <- extra[["week"]] * t + extra[["week^2"]] * (t^2)

  # 3. Gap effect
  gap_effect <- extra[["gap"]] * gap

  # This is if yprev is equal to 1 (the reference group)
  yprev_effect <- 0
  yprev_gap_effect <- 0

  # 4. Previous state effect
  yprev_coef_name <- paste0('yprev=', yprev)
  if (yprev_coef_name %in% names(extra)) {
      yprev_effect <- extra[[yprev_coef_name]]
  }

  yprev_gap_coef_name <- paste0('yprev=', yprev, ' * gap')
  if (yprev_gap_coef_name %in% names(extra)) {
      yprev_gap_effect <- extra[[yprev_gap_coef_name]] * gap
  }

  # Sum of all proportional effects
  lin_pred_prop <- tx_effect + baseline_effect + time_effect + gap_effect +
yprev_effect + yprev_gap_effect

  # 5. Non-proportional effect of time
  # From cppo=function(y) y, the effect for P(Y>=k) is k * week * gamma
  # The columns of the final lp correspond to k = 2, 3, ..., 8
  nonprop_gamma <- extra[['week x f(y)']]
  k_levels <- 2:8
  lin_pred_nonprop <- k_levels * t * nonprop_gamma
```

```r
  # 6. Final linear predictor (vector of length 7)
  # This is the sum of the proportional part (a scalar) and non-proportional part (a
vector)
  lp <- lin_pred_prop + lin_pred_nonprop

  return(lp)

}

formals(g)$extra <- extra
```

Function B (no impact on survival)

- This is a somewhat implausible scenario
- For LIQPLAT we hypothesize that ctDNA might lead to earlier deescalation, less toxicity and in turn better QoL
- The quality of life is thus mediated by bodily changes, which may impact survival as well
- For simulation purposes, we could handle a situation where the treatment does not affect survival with a conservative prior on the non-proportionality of the treatment effect

```r
g2 <- function(yprev, t, gap, X, parameter = 0, extra) {

  # X is a vector for one patient, containing baseline covariates and tx
  # yprev is the previous state (character)
  # t is the current week (numeric)
  # the gaps between measurements gap is assumed to be 1 for this simulation
  gap <- 1

  # 0. Treatment Effect
  # The treatment indicator 'tx' is the last element of the X vector.
  tx_value <- X[length(X)]
  tx_effect <- parameter * tx_value

  # 1. Baseline effect
  # X contains the non-treatment variables. We match its names to the 'extra' vector.
  X_nontx <- X[-length(X)] # We remove tx which is as the end of this vector
  baseline_coef_names <- names(X_nontx)

  baseline_effect <- sum(extra[baseline_coef_names] * X_nontx)

  # 2.  time effect (proportional part) from pol(week, 2)
  time_effect <- extra[["week"]] * t + extra[["week^2"]] * (t^2)

  # 3. Gap effect
  gap_effect <- extra[["gap"]] * gap

  # This is if yprev is equal to 1 (the reference group)
  yprev_effect <- 0
  yprev_gap_effect <- 0

  # 4. Previous state effect
  yprev_coef_name <- paste0('yprev=', yprev)
  if (yprev_coef_name %in% names(extra)) {
```

```
      yprev_effect <- extra[[yprev_coef_name]]
  }

  yprev_gap_coef_name <- paste0('yprev=', yprev, ' * gap')
  if (yprev_gap_coef_name %in% names(extra)) {
      yprev_gap_effect <- extra[[yprev_gap_coef_name]] * gap
  }

  # Sum of all proportional effects
  lin_pred_prop <- baseline_effect + time_effect + gap_effect + yprev_effect +
yprev_gap_effect

  # 5. Non-proportional effect of time
  # From cppo=function(y) y, the effect for P(Y>=k) is k * week * gamma
  # The columns of the final lp correspond to k = 2, 3, ..., 8
  nonprop_gamma <- extra[['week x f(y)']]
  k_levels <- 2:8
  lin_pred_nonprop <- k_levels * t * nonprop_gamma

  # 6. Final linear predictor (vector of length 7)
  # This is the sum of the proportional part (a scalar) and non-proportional part (a
vector)
  lp <- lin_pred_prop + lin_pred_nonprop + c(rep(tx_effect, 6), 0) # tx not applied to
last threshold

  return(lp)

}

formals(g2)$extra <- extra
```

Simulations

PO - OR 0.8

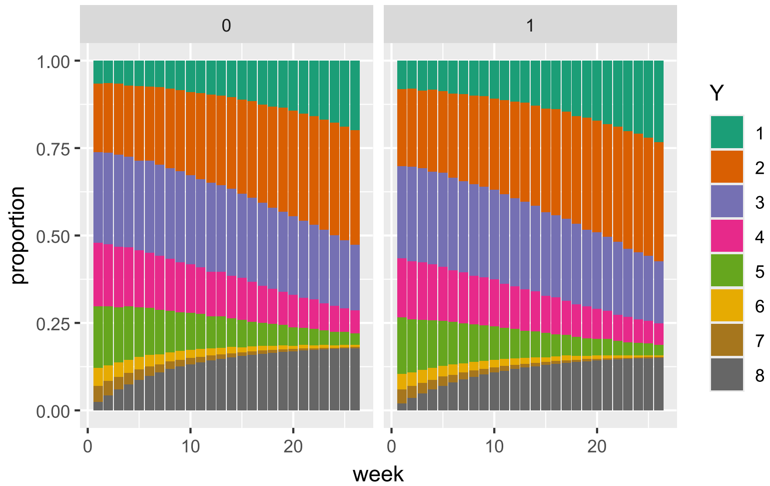
- Odds of 0.8 of moving to a higher (worse state)

```
# Run the simulation
carried_forward_08 <- simMarkovOrd(n = N,
                  y = levels(factor(na.omit(df)$y, levels = 1:8, ordered = TRUE)),
# 8 levels of the outcome
                  times = 1:26,
                  initial = init,
                  X = X,
                  absorb = 8,
                  intercepts = ints,
                  g = g,
                  parameter = log(0.8),
                  carry = TRUE)

not_carried_forward_08 <- simMarkovOrd(n = N,
                  y = levels(factor(na.omit(df)$y, levels = 1:8, ordered = TRUE)),
# 8 levels of the outcome
                  times = 1:26,
                  initial = init,
```

```
                     X = X,
                     absorb = '8',
                     intercepts = ints,
                     g = g,
                     parameter = log(0.8),
                     carry = FALSE)
```

Visualize empirical SOPs

```
carried_forward_08 |>
  ggplot(aes(x = time, fill = factor(y))) +
  geom_bar(position = "fill") +
  scale_fill_brewer(palette = "Dark2") +
  facet_wrap(~ tx) +
  labs(x = "week", y = "proportion", fill = "Y")
```



Check if OR of 0.8 is recovered

```
sample_08 <- not_carried_forward_08 |>
    select(id, time, y, yprev, tx) |>
    left_join(sim_baseline_data |> select(-yprev), by = c("id" = "pat_id"))

h <- vglm(ordered(y) ~ yprev + poly(time, 2) + ecog_fstcnt + diagnosis + tx,
                cumulative(reverse=TRUE, parallel=FALSE ~ pol(time, 2)),
                data = sample_08 |> filter(id < 10000))

h
```

```

Call:
vglm(formula = ordered(y) ~ yprev + poly(time, 2) + ecog_fstcnt +
    diagnosis + tx, family = cumulative(reverse = TRUE, parallel = FALSE ~
    pol(time, 2)), data = filter(sample_08, id < 10000))


Coefficients:
   (Intercept):1     (Intercept):2     (Intercept):3     (Intercept):4
```

```
       2.4486049        0.6475968       -0.7187021       -1.7096675
    (Intercept):5    (Intercept):6    (Intercept):7           yprev2
      -3.1541962       -3.8899322       -5.2161251       -0.3139428
          yprev3           yprev4           yprev5           yprev6
      -0.6802082        0.3169778       -0.7286248       -0.6664955
          yprev7   poly(time, 2)1   poly(time, 2)2     ecog_fstcnt2
       0.4103984     -253.4732023      -12.8388625        1.4173312
ecog_fstcnt3plus       diagnosis2       diagnosis3       diagnosis4
       2.6517694       -0.2221567       -0.8070174       -0.4815620
      diagnosis5               tx
      -0.1866917       -0.2003369

Degrees of Freedom: 1610161 Total; 1610139 Residual
Residual deviance: 714647.4
Log-likelihood: -357323.7
```

Calculate the SOPs

```
sops <- carried_forward_08 |>
  group_by(time, tx) |>
  count(y) |>
  reframe(
    y = y,
    sop = n / sum(n))
```

Calculate the difference in weeks with >= Good QoL

```
# Difference in mean days with excellent, very good or good quality of life
weeks_good_qol <- sops |>
  filter(y == 1 | y == 2 | y == 3) |>
  group_by(tx) |>
  summarise(mean_weeks = sum(sop))

weeks_good_qol
```

```
# A tibble: 2 × 2
     tx mean_weeks
  <dbl>      <dbl>
1     0       16.0
2     1       17.1
```

```
weeks_bad_qol <- sops |>
  filter(y == 7 | y == 6 | y == 5 | y == 8) |>
  group_by(tx) |>
  summarise(mean_weeks = sum(sop))

weeks_bad_qol
```

```
# A tibble: 2 × 2
     tx mean_weeks
  <dbl>      <dbl>
```

```
1     0       6.83
2     1       5.89
```

In this population an OR of 0.8 for a proportional treatment effect leads to a 1.1 week increase in the expected time spent with a quality of life in the highest three categories. The difference in weeks spent with bad quality of life or death is −0.9.

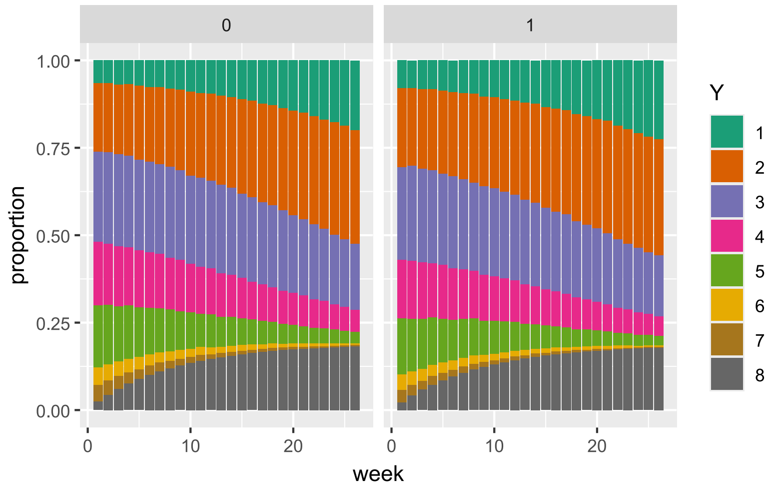
nonPO - OR 0.8

- Odds of 0.8 of moving to a higher (worse state), except for the absorbing state death, which has OR of 1.

```
# Run the simulation
carried_forward_nonPO_08 <- simMarkovOrd(n = N,
                  y = levels(factor(na.omit(df)$y, levels = 1:8, ordered = TRUE)), # 8
levels of the outcome
                  times = 1:26,
                  initial = init,
                  X = X,
                  absorb = 8,
                  intercepts = ints,
                  g = g2,
                  parameter = log(0.8),
                  carry = TRUE)

not_carried_forward_nonPO_08 <- simMarkovOrd(n = N,
                  y = levels(factor(na.omit(df)$y, levels = 1:8, ordered = TRUE)),
                  times = 1:26,
                  initial = init,
                  X = X,
                  absorb = '8',
                  intercepts = ints,
                  g = g2,
                  parameter = log(0.8),
                  carry = FALSE)
```

Visualize empirical SOPs

```
carried_forward_nonPO_08 |>
  ggplot(aes(x = time, fill = factor(y))) +
  geom_bar(position = "fill") +
  scale_fill_brewer(palette = "Dark2") +
  facet_wrap(~ tx) +
  labs(x = "week", y = "proportion", fill = "Y")
```



Examine OR

```
sample_nonPO_08 <- not_carried_forward_nonPO_08 |>
    select(id, time, y, yprev, tx) |>
    left_join(sim_baseline_data |> select(-yprev), by = c("id" = "pat_id"))

h_nonPO <- vglm(ordered(y) ~ yprev + poly(time, 2) + ecog_fstcnt + diagnosis + tx,
                cumulative(reverse=TRUE, parallel=FALSE ~ pol(time, 2)+tx),
                data=sample_nonPO_08 |> filter(id < 10000))

h_nonPO
```

```

Call:
vglm(formula = ordered(y) ~ yprev + poly(time, 2) + ecog_fstcnt +
    diagnosis + tx, family = cumulative(reverse = TRUE, parallel = FALSE ~
    pol(time, 2) + tx), data = filter(sample_nonPO_08, id < 10000))


Coefficients:
   (Intercept):1     (Intercept):2     (Intercept):3     (Intercept):4
    2.480976e+00      6.676108e-01     -7.028244e-01     -1.703108e+00
   (Intercept):5     (Intercept):6     (Intercept):7            yprev2
   -3.152536e+00     -3.905242e+00     -5.205223e+00     -3.022903e-01
          yprev3            yprev4            yprev5            yprev6
   -6.991728e-01      3.048288e-01     -7.632921e-01     -6.888675e-01
          yprev7    poly(time, 2)1    poly(time, 2)2      ecog_fstcnt2
    4.081865e-01     -2.576295e+02     -1.506781e+01      1.447871e+00
ecog_fstcnt3plus        diagnosis2        diagnosis3        diagnosis4
    2.667850e+00     -2.239217e-01     -7.905600e-01     -4.928289e-01
      diagnosis5              tx:1              tx:2              tx:3
   -1.985642e-01     -2.092191e-01     -2.106215e-01     -2.173185e-01
            tx:4              tx:5              tx:6              tx:7
   -1.972127e-01     -2.127526e-01     -1.834810e-01      9.834382e-04

Degrees of Freedom: 1593151 Total; 1593123 Residual
Residual deviance: 705606.1
Log-likelihood: -352803.1
```

VGAM estimated separate effects for each cumulative threshold. Correctly estimated 0.8 for all but threshold 7, which is estimated as 1.

Calculate the SOPs

```
sops_nonPO <- carried_forward_nonPO_08 |>
  group_by(time, tx) |>
  count(y) |>
  reframe(
    y = y,
    sop = n / sum(n))
```

Calculate the difference in weeks with >= Good QoL

```
# Difference in mean days with excellent, very good or good quality of life
weeks_good_qol_nonPO <- sops_nonPO |>
  filter(y == 1 | y == 2 | y == 3) |>
  group_by(tx) |>
  summarise(mean_weeks = sum(sop))

weeks_good_qol_nonPO
```

```
# A tibble: 2 × 2
     tx mean_weeks
  <dbl>      <dbl>
1     0       15.9
2     1       16.8
```

```
weeks_bad_qol_nonPO <- sops_nonPO |>
  filter(y == 7 | y == 6 | y == 5) |>
  group_by(tx) |>
  summarise(mean_weeks = sum(sop))

weeks_bad_qol_nonPO
```

```
# A tibble: 2 × 2
     tx mean_weeks
  <dbl>      <dbl>
1     0       3.35
2     1       2.85
```

If the treatment has no effect on mortality, the effect is reduced from a gain of 1.1 weeks week >= Good QoL to a gain of 0.9 weeks.

nonPO - OR 0.3

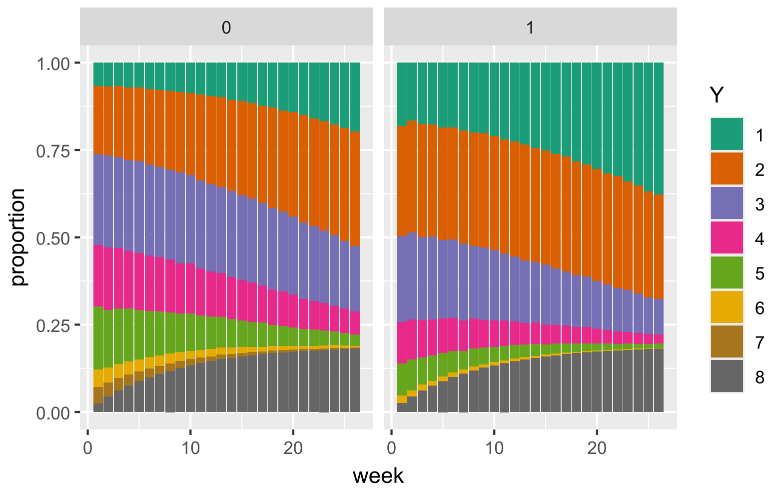
- Odds Ratio of 0.3 of moving to a higher (worse state), except for the absorbing state death

```
# Run the simulation
carried_forward_nonPO <- simMarkovOrd(n = N,
                  y = levels(factor(na.omit(df)$y, levels = 1:8, ordered = TRUE)), # 8
levels of the outcome
                  times = 1:26,
```

```
                      initial = init,
                      X = X,
                      absorb = 8,
                      intercepts = ints,
                      g = g2,
                      parameter = log(0.3),
                      carry = TRUE)

not_carried_forward_nonPO <- simMarkovOrd(n = N,
                      y = levels(factor(na.omit(df)$y, levels = 1:8, ordered = TRUE)),
                      times = 1:26,
                      initial = init,
                      X = X,
                      absorb = '8',
                      intercepts = ints,
                      g = g2,
                      parameter = log(0.3),
                      carry = FALSE)
```

Visualize empirical SOPs

```
carried_forward_nonPO |>
  ggplot(aes(x = time, fill = factor(y))) +
  geom_bar(position = "fill") +
  scale_fill_brewer(palette = "Dark2") +
  facet_wrap(~ tx) +
  labs(x = "week", y = "proportion", fill = "Y")
```



Examine OR

```
sample_nonPO <- not_carried_forward_nonPO |>
    select(id, time, y, yprev, tx) |>
    left_join(sim_baseline_data |> select(-yprev), by = c("id" = "pat_id"))

h_nonPO <- vglm(ordered(y) ~ yprev + poly(time, 2) + ecog_fstcnt + diagnosis + tx,
                cumulative(reverse=TRUE, parallel=FALSE ~ pol(time, 2)+tx),
                data=sample_nonPO |> filter(id < 10000))
```

```
h_nonPO
```

```

Call:
vglm(formula = ordered(y) ~ yprev + poly(time, 2) + ecog_fstcnt +
    diagnosis + tx, family = cumulative(reverse = TRUE, parallel = FALSE ~
    pol(time, 2) + tx), data = filter(sample_nonPO, id < 10000))


Coefficients:
   (Intercept):1    (Intercept):2    (Intercept):3    (Intercept):4
      2.48242501       0.67137199      -0.70344212      -1.69371957
   (Intercept):5    (Intercept):6    (Intercept):7           yprev2
     -3.11909168      -3.84478920      -5.12034689      -0.32660755
          yprev3           yprev4           yprev5           yprev6
     -0.69240959       0.31791544      -0.76922730      -0.69807801
          yprev7   poly(time, 2)1   poly(time, 2)2     ecog_fstcnt2
      0.44397232    -248.06909040     -14.65876297       1.42488322
ecog_fstcnt3plus       diagnosis2       diagnosis3       diagnosis4
      2.61999630      -0.20923312      -0.79793633      -0.47422717
      diagnosis5             tx:1             tx:2             tx:3
     -0.19276486      -1.17495139      -1.14264693      -1.13258340
            tx:4             tx:5             tx:6             tx:7
     -1.12765253      -1.18216569      -1.19505095      -0.03595726

Degrees of Freedom: 1585563 Total; 1585535 Residual
Residual deviance: 665287.9
Log-likelihood: -332644
```

VGAM estimated separate effects for each cumulative threshold. Correctly estimated 0.3 for all but threshold 7, which is estimated as 1.

Calculate the SOPs

In those that survive > 26 weeks (primary estimand).

```
sops_nonPO <- carried_forward_nonPO |>
  group_by(id) |>
  filter(all(y != 8)) |>
  ungroup() |>
  group_by(time, tx) |>
  count(y) |>
  reframe(
    y = y,
    sop = n / sum(n))
```

Calculate the difference in weeks with >= Good QoL

```
# Difference in mean days with excellent, very good or good quality of life
weeks_good_qol_nonPO <- sops_nonPO |>
  filter(y == 1 | y == 2 | y == 3) |>
  group_by(tx) |>
  summarise(mean_weeks = sum(sop))

```

```
weeks_good_qol_nonPO$mean_weeks[2] - weeks_good_qol_nonPO$mean_weeks[1]
```

```
[1] 4.035204
```

```
weeks_good_qol_nonPO
```

```
# A tibble: 2 × 2
     tx mean_weeks
  <dbl>      <dbl>
1     0       18.8
2     1       22.8
```

```
weeks_bad_qol_nonPO <- sops_nonPO |>
  filter(y == 7 | y == 6 | y == 5) |>
  group_by(tx) |>
  summarise(mean_weeks = sum(sop))

weeks_bad_qol_nonPO
```

```
# A tibble: 2 × 2
     tx mean_weeks
  <dbl>      <dbl>
1     0       3.59
2     1       1.31
```

If the treatment has no effect on mortality, the effect of an OR of 0.3 is a gain of 3 weeks with good QoL.

nonOR - OR 3.33

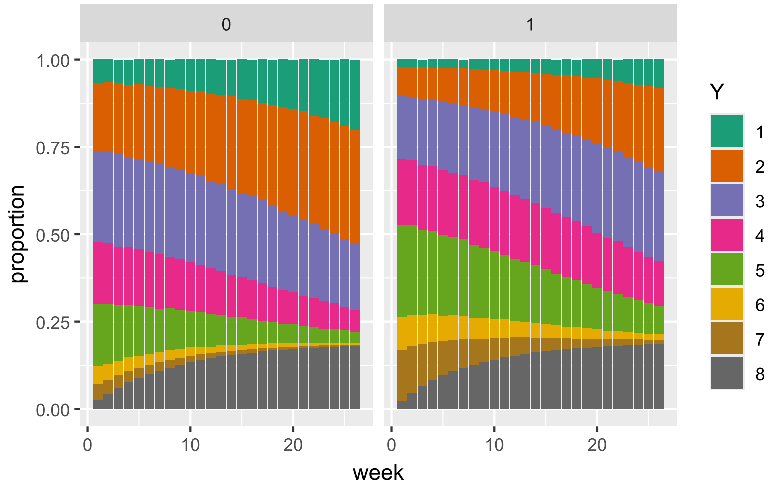
- Odds Ratio of 3.33 of moving to a higher (worse state), except for the absorbing state death

```
# Run the simulation
carried_forward_nonPO <- simMarkovOrd(n = N,
                  y = levels(factor(na.omit(df)$y, levels = 1:8, ordered = TRUE)), # 8
levels of the outcome
                  times = 1:26,
                  initial = init,
                  X = X,
                  absorb = 8,
                  intercepts = ints,
                  g = g2,
                  parameter = log(3.33),
                  carry = TRUE)

not_carried_forward_nonPO <- simMarkovOrd(n = N,
                  y = levels(factor(na.omit(df)$y, levels = 1:8, ordered = TRUE)),
                  times = 1:26,
                  initial = init,
                  X = X,
                  absorb = '8',
                  intercepts = ints,
```

```
                   g = g2,
                   parameter = log(3.33),
                   carry = FALSE)
```

Visualize empirical SOPs

```
carried_forward_nonPO |>
  ggplot(aes(x = time, fill = factor(y))) +
  geom_bar(position = "fill") +
  scale_fill_brewer(palette = "Dark2") +
  facet_wrap(~ tx) +
  labs(x = "week", y = "proportion", fill = "Y")
```



Examine OR

```
sample_nonPO <- not_carried_forward_nonPO |>
    select(id, time, y, yprev, tx) |>
    left_join(sim_baseline_data |> select(-yprev), by = c("id" = "pat_id"))

h_nonPO <- vglm(ordered(y) ~ yprev + poly(time, 2) + ecog_fstcnt + diagnosis + tx,
                cumulative(reverse=TRUE, parallel=FALSE ~ pol(time, 2)+tx),
                data=sample_nonPO |> filter(id < 10000))

h_nonPO
```

```

Call:
vglm(formula = ordered(y) ~ yprev + poly(time, 2) + ecog_fstcnt +
    diagnosis + tx, family = cumulative(reverse = TRUE, parallel = FALSE ~
    pol(time, 2) + tx), data = filter(sample_nonPO, id < 10000))


Coefficients:
   (Intercept):1     (Intercept):2     (Intercept):3     (Intercept):4
      2.50441180        0.70377609       -0.67319699       -1.68092180
   (Intercept):5     (Intercept):6     (Intercept):7            yprev2
     -3.13617305       -3.88397406       -5.21756957       -0.32789355
```

```
         yprev3          yprev4          yprev5          yprev6
    -0.71750849      0.28850795     -0.75946812     -0.65754008
         yprev7  poly(time, 2)1  poly(time, 2)2     ecog_fstcnt2
     0.42423508   -264.22300800    -13.25192601      1.42871102
ecog_fstcnt3plus     diagnosis2      diagnosis3       diagnosis4
     2.65557579     -0.22128278     -0.78833552     -0.51287766
     diagnosis5            tx:1            tx:2             tx:3
    -0.20898707      1.19922226      1.16166104       1.15036005
           tx:4            tx:5            tx:6             tx:7
     1.15207401      1.15634652      1.15703038       0.08058002

Degrees of Freedom: 1584086 Total; 1584058 Residual
Residual deviance: 731194.5
Log-likelihood: -365597.3
```

VGAM estimated separate effects for each cumulative threshold. Correctly estimated 3.33 for all but threshold 7, which is estimated as 1.

Calculate the SOPs

In those that survive > 26 weeks (primary estimand).

```
sops_nonPO <- carried_forward_nonPO |>
  group_by(id) |>
  filter(all(y != 8)) |>
  ungroup() |>
  group_by(time, tx) |>
  count(y) |>
  reframe(
    y = y,
    sop = n / sum(n))
```

Calculate the difference in weeks with >= Good QoL

```
# Difference in mean days with excellent, very good or good quality of life
weeks_good_qol_nonPO <- sops_nonPO |>
  filter(y == 1 | y == 2 | y == 3) |>
  group_by(tx) |>
  summarise(mean_weeks = sum(sop))

weeks_good_qol_nonPO
```

```
# A tibble: 2 × 2
     tx mean_weeks
  <dbl>      <dbl>
1     0       18.8
2     1       12.9
```

```
weeks_good_qol_nonPO$mean_weeks[2] - weeks_good_qol_nonPO$mean_weeks[1]
```

```
[1] -5.899842
```

```
weeks_bad_qol_nonPO <- sops_nonPO |>
  filter(y == 7 | y == 6 | y == 5) |>
  group_by(tx) |>
  summarise(mean_weeks = sum(sop))

weeks_bad_qol_nonPO
```

```
# A tibble: 2 × 2
     tx mean_weeks
  <dbl>      <dbl>
1     0       3.57
2     1       7.85
```

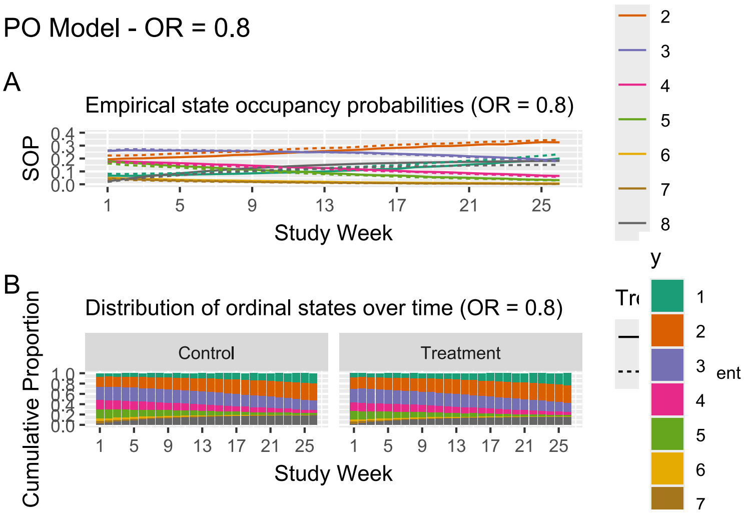
If the treatment has no effect on mortality, the effect of an OR of 3.3 is a loss of 6 weeks with good QoL.

Create and export images for OR = 0.8

State Occupancy Probabilities

```
source(here("functions", "carry-death-forward.R"))

sample_cf_08 <- sample_08 |>
  carry_death_forward() |>
  mutate(
    tx = factor(tx, levels = c(0, 1), labels = c("Control", "Treatment"))
  )

sops_line_08 <- sample_cf_08 |>
  group_by(time, tx) |>
  count(y) |>
  mutate(
    total = sum(n),
    sop   = n / total,
    tx = factor(tx)) |>
  ungroup() |>
  ggplot(aes(x = time, y = sop, color = y)) +
  geom_line(aes(linetype = tx)) +
  scale_color_brewer(palette = "Dark2") +
  scale_x_continuous(breaks = seq(1, 26, by=4)) +
  scale_y_continuous(breaks = seq(0, 1, by=0.1)) +
  coord_cartesian(ylim = c(0, 0.4)) +
  labs(x = "Study Week",
   y = "SOP",
   subtitle = "Empirical state occupancy probabilities (OR = 0.8)",
   linetype = "Treatment",
   color = "State")

sops_stacked_08 <- sample_cf_08 |>
  ggplot(aes(x = time, fill = y)) +
  geom_bar(position = "fill")  +
  scale_fill_brewer(palette = "Dark2") +
  scale_x_continuous(breaks = seq(1, 26, by=4)) +
  scale_y_continuous(breaks = seq(0, 1, by=0.2)) +
  facet_wrap(~tx) +
```

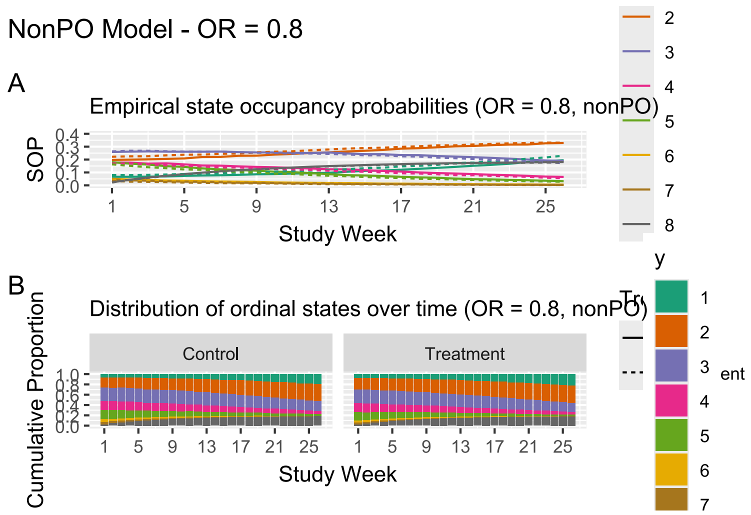
```
  labs(x = "Study Week",
   y = "Cumulative Proportion",
   subtitle = "Distribution of ordinal states over time (OR = 0.8)",
   color = "State")

sops_line_08 / sops_stacked_08 +
  plot_annotation(tag_level = 'A',
                  title = "PO Model - OR = 0.8")
```




```
ggsave(filename = here("img", "fig-sim-data-sop.svg"), width = 12, height = 10)
```

```
# State Occupancy Probabilities - nonPO model
sample_nonPO_cf_08 <- sample_nonPO_08 |>
  carry_death_forward() |>
  mutate(
    tx = factor(tx, levels = c(0, 1), labels = c("Control", "Treatment"))
  )

sops_line_nonPO_08 <- sample_nonPO_cf_08 |>
  group_by(time, tx) |>
  count(y) |>
  mutate(
    total = sum(n),
    sop   = n / total,
    tx = factor(tx)) |>
  ungroup() |>
  ggplot(aes(x = time, y = sop, color = y)) +
  geom_line(aes(linetype = tx)) +
  scale_color_brewer(palette = "Dark2") +
  scale_x_continuous(breaks = seq(1, 26, by=4)) +
  scale_y_continuous(breaks = seq(0, 1, by=0.1)) +
  coord_cartesian(ylim = c(0, 0.4)) +
  labs(x = "Study Week",
   y = "SOP",
   subtitle = "Empirical state occupancy probabilities (OR = 0.8, nonPO)",
   linetype = "Treatment",
   color = "State")
```

```
sops_stacked_nonPO_08 <- sample_nonPO_cf_08 |>
  ggplot(aes(x = time, fill = y)) +
  geom_bar(position = "fill")  +
  scale_fill_brewer(palette = "Dark2") +
  scale_x_continuous(breaks = seq(1, 26, by=4)) +
  scale_y_continuous(breaks = seq(0, 1, by=0.2)) +
  facet_wrap(~tx) +
  labs(x = "Study Week",
   y = "Cumulative Proportion",
   subtitle = "Distribution of ordinal states over time (OR = 0.8, nonPO)",
   color = "State")

sops_line_nonPO_08 / sops_stacked_nonPO_08 +
  plot_annotation(tag_level = 'A',
                  title = "NonPO Model - OR = 0.8")
```




```
ggsave(filename = here("img", "fig-sim-data-sop-nonPO.svg"), width = 12, height = 10)
```

Individual trajectories

```
set.seed(4)
ids_08 <- sample_08$id |> unique() |> sample(20)

fig_tile_random_08 <- sample_08 |>
  filter(id %in% ids_08) |>
  mutate(id = as.factor(as.numeric(as.factor((id))))) |>
  ggplot(aes(y = id, x = time)) +
  geom_tile(mapping = aes(fill = y), width = 0.95, height = 0.4) +
  scale_fill_brewer(palette = "Dark2") +
  scale_x_continuous(breaks = 1:26) +
  labs(title = "Individual Trajectories - OR = 0.8",
       y = "Patient ID",
       x = "Study Week",
       fill = "State")

ggsave(filename = here("img", "fig-sim-tile.svg"), width = 12, height = 6)
```

Export simulataed datasets

```
write_parquet(sample_08, here("1-qol", "sim-data", "sim_data_po.parquet"))
write_parquet(carried_forward_08, here("1-qol", "sim-data",
"sim_data_po_forward.parquet"))
write_parquet(sample_nonPO_08, here("1-qol", "sim-data", "sim_data_nonpo.parquet"))
```